Phase 0: Foundations

A visual, step-by-step guide - what it builds, what it leaves alone, what you will see

In one sentence

Phase 0 builds the **plumbing** for everything that comes after it: a per-school on/off switch, a phone-number cleaner, safe logging, Nairobi time, a place to run background jobs, and a health check. It also fixes the admin dashboard's wrong counts.

The important bit

No parent will see anything yet, and no school console screen changes. The only visible change in a browser is that the admin dashboard counts become correct.

Think of it like building a house

You do not see pipes when you walk into a finished house, but nothing else works without them. That is Phase 0.

What Phase 0 builds

- Per-school feature switches
- Phone-number cleaner
- Safe logging (masked numbers)
- Nairobi time helper
- Redis queue for new jobs
- Health check command
- Config key names
- Dashboard count fix

What it leaves alone

- Driver app and Guardian app
- Push notifications
- Every existing endpoint
- users/all-schools (only written up)
- Server time zone (stays UTC)
- The .env file (never read)

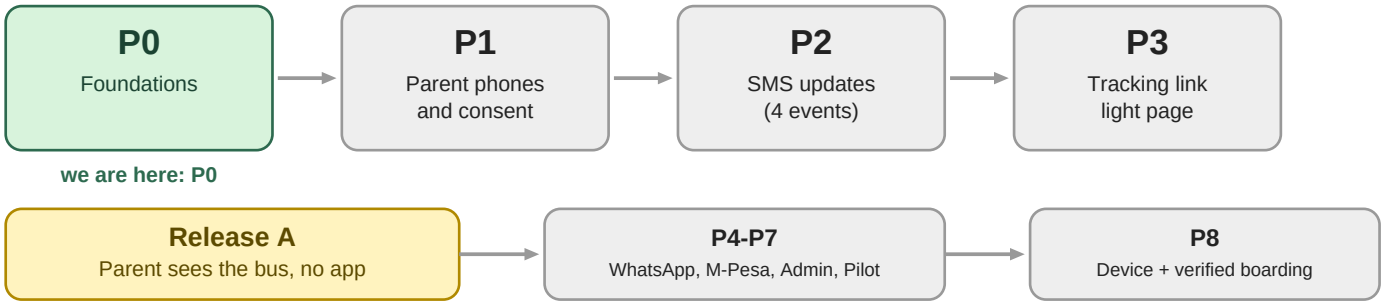
What you will see

- Correct admin dashboard counts
- Switch list: all six OFF
- Health check output
- 30+ passing phone tests
- Before/after test results
- A report under 500 words

Where Phase 0 sits

The road to 'parent sees the bus', and what is old versus new

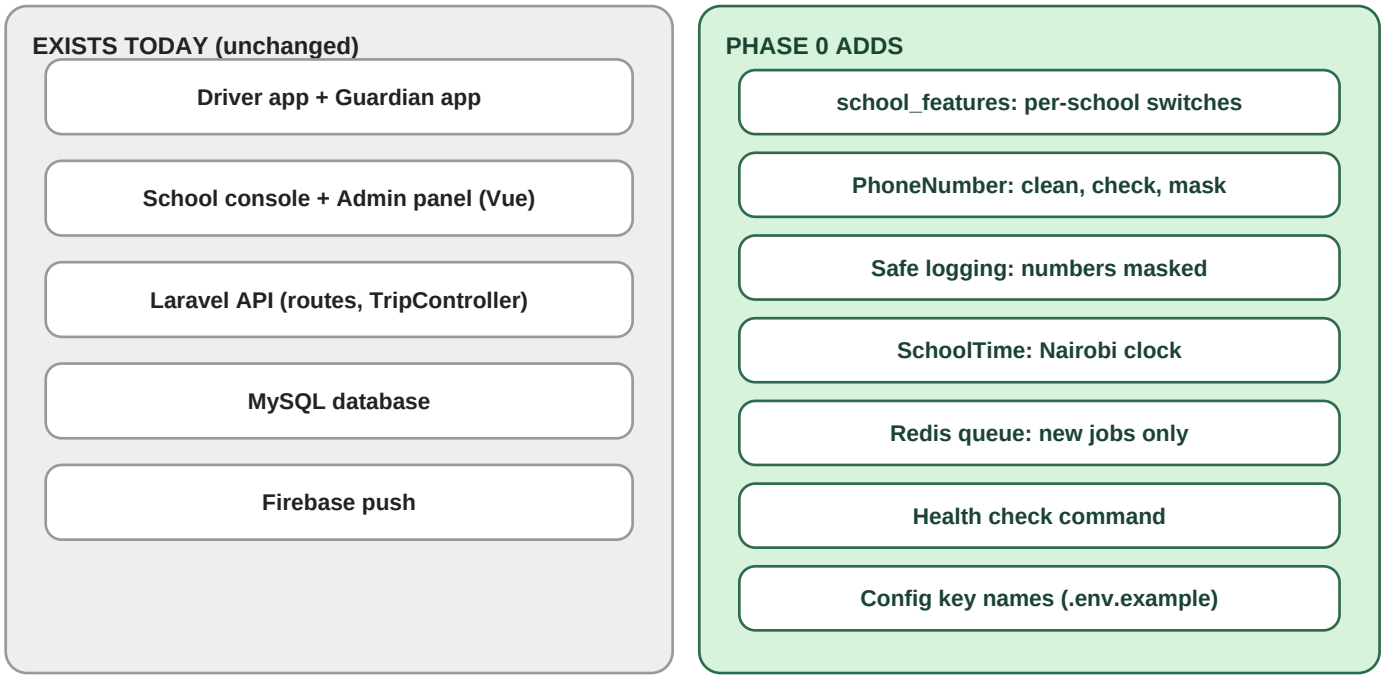
The road ahead



Why Phase 0 comes first

Later phase	Needs from Phase 0
P1 Parent phones	Phone helper, feature switches
P2 SMS updates	Queue, Nairobi time, masked logs
P3 Tracking link	Feature switches
Every phase	Health check

Picture of the system: old versus new



Amber = one small fix

The admin dashboard counts the wrong roles. Phase 0 edits just **AdminDashboardController** so the counts are right. Grey = unchanged, green = new.

The steps, one by one

Phase 0 has nine small steps. Do them in this order.

0 Baseline commit and test run

WHAT IT DOES

Saves a first snapshot of the project (without the .env secrets file) so every later change can be reviewed. Then runs the existing tests to record the starting score.

YOU WILL SEE

A before-and-after test result: pass and fail counts.

1 Dashboard count fix

WHAT IT DOES

The admin dashboard counts drivers, parents and students using the wrong role numbers. One file, AdminDashboardController, is corrected.

YOU WILL SEE

Correct counts on the admin dashboard (table on page 6).

2 school_features: the switches

WHAT IT DOES

A table of per-school on/off switches: parent_mode, sms_updates, tracking_links, whatsapp, mpesa, verified_boarding. All OFF by default. Only admins can flip them, and every flip is recorded.

YOU WILL SEE

An admin-only API to read and flip switches. No screen yet.

3 PhoneNumber helper

WHAT IT DOES

Turns any Kenyan number format into one standard +254 form, rejects bad numbers, and hides the middle digits when shown. Existing numbers in the database are not rewritten.

YOU WILL SEE

The examples table on page 5, backed by 30+ tests.

4 Safe logging

WHAT IT DOES

New outgoing web calls log their activity with phone numbers masked and secrets removed. Old code that logs device tokens is not edited, only listed for you.

YOU WILL SEE

Log lines like the example on page 5.

The steps, continued

Steps 5 to 8, then the order rule

5 SchoolTime: Nairobi clock

WHAT IT DOES

One place that gives Nairobi time to new code. The server itself stays on UTC, so nothing existing shifts.

YOU WILL SEE

Correct school-day boundaries in later phases.

6 Redis queue for new jobs

WHAT IT DOES

New background jobs (SMS, tracking links) get a real queue. Existing code keeps today's behaviour and runs immediately.

YOU WILL SEE

A supervisor config example in the docs folder.

7 Config key names

WHAT IT DOES

Names for the WhatsApp, SMS, Daraja (M-Pesa) and short-link settings, with no values filled in.

YOU WILL SEE

New lines in .env.example.

8 Health check command

WHAT IT DOES

One command that says whether Redis, the queue worker and the scheduler are alive, and names the broken one if not.

YOU WILL SEE

Output like the example on page 6.

The flow in one line

snapshot

Step 0 comes first so that every later change is measured against a known starting point. Steps 1 to 8 are small, separate and additive, so a problem in one cannot break the others.

What it looks like (expected)

Mock-ups showing the intended result, not real output

The switches in school_features

School 12 "Example Primary"

parent_mode	OFF	sms_updates	OFF
tracking_links	OFF	whatsapp	OFF
mpesa	OFF	verified_boarding	OFF

Only an admin can change these. Schools cannot. **Turning a switch off returns that school to exactly today's behaviour.**

Phone numbers: what you type, what is kept, what is shown

You type	Stored and used	Shown on screens and logs
0712345678	+254712345678	+254712***678
0112345678	+254112345678	+254112***678
712345678	+254712345678	+254712***678
254712345678	+254712345678	+254712***678
+254 712 345 678	+254712345678	+254712***678
071234567 (too short)	**REJECTED**	n/a
0612345678 (not mobile)	**REJECTED**	n/a

Existing numbers in the users table are **not rewritten**. The helper cleans them when read. Final accepted ranges are confirmed by the 30+ tests.

Safe logging: before and after

UNSAFE - must never appear

POST sms/send to=+254712345678 key=sk_live_AbC123

SAFE - what Phase 0 produces

POST sms/send to=+254712***678 key=[hidden]

Old code that logs device tokens and ticket numbers is **not edited**. Phase 0 only lists those lines for you.

Health check, dashboard fix, queue

What you will see and what changes

PHASE 0

Health check command (name is a proposal)

```
$ php artisan health:check

Redis ....... OK   (3 ms)
Queue worker  OK   (12 s ago)
Scheduler .... OK   (41 s ago)
Time ... Nairobi 07:14 / UTC 04:14
All checks passed.
```

Everything healthy

```
$ php artisan health:check

Redis ....... OK
Queue worker  FAIL
 no job in 15 min - is worker on?
Scheduler .... OK
1 problem found.
```

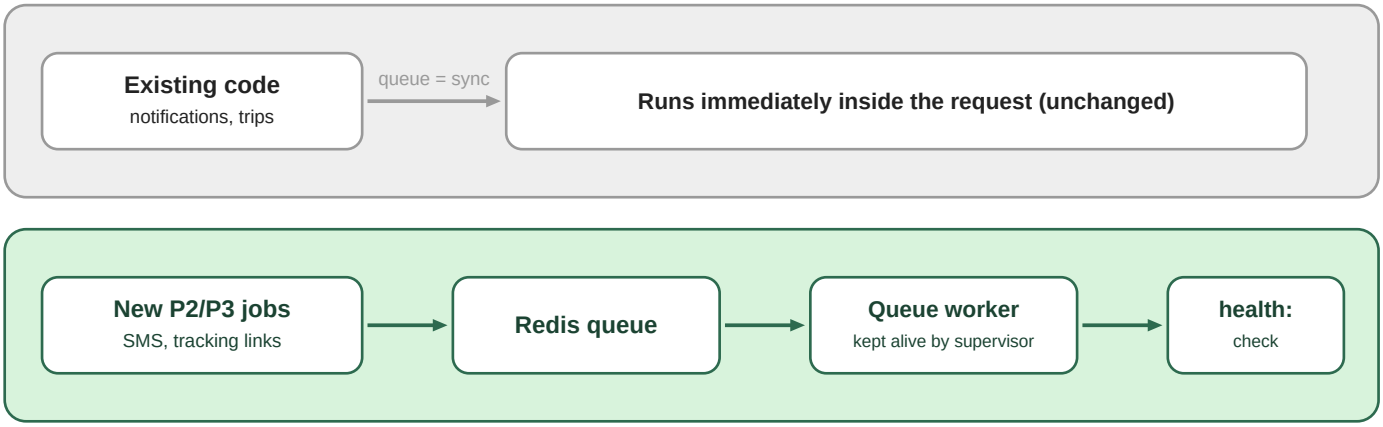
Something wrong: it names the problem

Dashboard fix: only these numbers change

Dashboard label	Role number today	Role number after Phase 0
Schools	2 (correct)	2
Drivers	**2 (wrong)**	**3**
Parents	**3 (wrong)**	**4**
Guardians	5 (correct)	5
Students	**4 (wrong)**	**6**

No other output changes.

Queue: what changes and what does not



Safe by design

Only **new** jobs name the Redis connection. Everything that exists today keeps running exactly as before, so turning the queue on cannot change old behaviour.

Boundaries, files and sign-off

What Phase 0 does not do, and how to check it yourself

What Phase 0 does NOT do

- No change to the driver app, guardian app, push notifications or any existing endpoint.
- GET users/all-schools is not touched. Both apps call it. Phase 0 only writes up what it exposes and one safe fix for later.
- No messages sent, no parent screens, no tracking page, no M-Pesa.
- No change to the server time zone (stays UTC). The .env file is never read.

Files Phase 0 is expected to touch (to be confirmed in the plan step)

NEW

- Migration and model: school_features
- PhoneNumber and SchoolTime helpers
- Health check command
- Log-masking step for web calls
- Supervisor config example (docs)
- Tests for each of the above

EDITED (small, additive)

- AdminDashboardController.php (count fix)
- routes/api.php (one admin-only group)
- Console/Kernel.php (one heartbeat)
- AppServiceProvider.php (registers the web-call step)
- .env.example (key names only)

Check it yourself in 5 minutes

1. Open the admin dashboard: driver, parent and student counts match real users.
2. Ask for one school's switch list through the admin endpoint: all six read OFF.
3. Run the health command: all lines OK, or it names the broken one.
4. Look at the test report: all 30+ phone examples pass.
5. Compare test results before and after: same or better.

Phase 0 is done when

- [ ] Baseline snapshot exists, .env not included
- [ ] Dashboard counts correct, covered by tests
- [ ] 30+ phone tests pass
- [ ] Health command works and reports a failure correctly
- [ ] Report under 500 words delivered to you
- [ ] Full test results recorded before and after
- [ ] Six switches exist: default off, admin-only, audited
- [ ] Logs show masked numbers only
- [ ] No existing endpoint or app behaviour changed